EFELIA MIAI Cluster IA

GRENOBLE INP UGA Formation Pro

INSTITUT D'INGENIERIE ET DE MANAGEMENT

Data Quality



Smart Analytics for Big Data

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Program

Data Handling

Principle of this lesson

Content of the tutorial files and help

Data Quality

A typology of industrial data?

The 6 pillars of data quality

Data Vizu

Visualization

What? Why? How?

References

Context and Issues

- Data Types
- Pilars of the diagnostic of the industrial data quality

Data Handling

Principle of this lesson

Principle of this lesson

Material

- Some slides to present the lesson
- Tutorial webpage on data operations (+ *associated* `.Rmd`)
  - something is missing?:
    - Practical Data Science with R
    - https://stackoverflow.com/questions/tagged/r
- Applications

Principle of this lesson

Teaching Data Handling and Quality diagnostic

Activities

- **First level:**
  Read, understand & reproduce and modify the *example: BMI* [1] *analysis of Star Wars Characters in the tutorial*

[1] Body Mass Index

- **Intermediate level**:

  Choose a dataset and perform descriptive analysis of the data

  - Precise your scientific/business question
  - Develop CRISP-DM short description of the project
  - Import the new database
  - Analyse the data to answer your question
  - Produce a dynamic document

- **Advanced level**:

  Choose a dataset and perform and Precise your scientific/business question

  - data quality diagnostic
    - present the raw and prepared datasets (qualify your data)
    - present a descriptive analysis of the data (summary statistic table)
  - plan your ML/DL process and present the data preparation you will target

Content of the tutorial files and help

Data input and output

Inputs

- Paths with R
- Read
- Best Practices in Preparing Data Files for Importing into R

Output

- Displaying data
- Save data

Data management

- Structure and metadata
- Access variables from a dataset
- Names of variables and their types
- Values of variables in a dataset
- Display the variables values
- Indexing variables

Derived variables and data manipulation

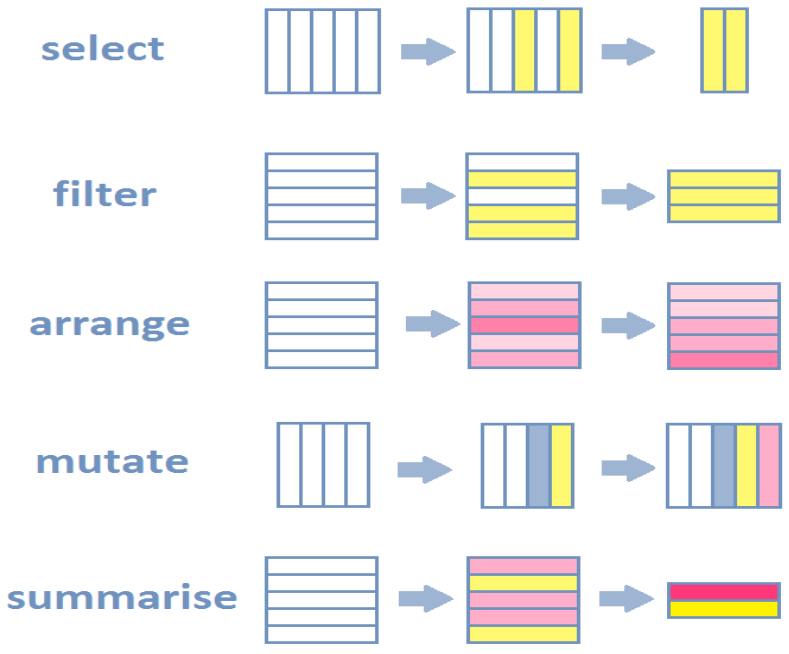


Figure 1: Dplyr data manipulation

- Data manipulation
  - `select`, to select variables (columns) of the data table
  - `filter`, to filter the individuals (rows) of the data table
  - `arrange`, to rearrange the data array according to the order of one or more variables
  - `mutate`, to create and add new variables (columns) to the dataset
  - `summarize`, to summarize several values into one

Application:

Ex: IMC of Star Wars characters

Usual Data Operations

- Conditional variable transformation
- Chaining
- Rename variables in a dataset
- Keep unique values
- Identify duplicated values

String variables

- Create string variables from numeric variables
- Create categorical variables from continuous variables
- Recode a categorical variable
- Create a categorical variable using logic
- Extract characters from string variables
- Length of string variables
- Concatenate string variables
- Find strings within string variables
- Replace strings within string variables
- Split strings into multiple strings
- Remove spaces around string variables
- Convert strings from upper to lower case

More functions

- Create lagged variable
- Factors
- Formatting values of variables
- Long and Wide formats
- Concatenate and stack datasets

Data Quality

Why industrial data quality?

Industrial environments (IoT, SCADA, MES, ERP) produce massive volumes of data.

But this data is often:

- incomplete
- noisy
- inconsistent
- time-shifted
  Without reliable data quality, industrial decisions are biased.

Purpose of the diagnostic: identify, quantify and understand data defects.

A typology of industrial data?

A typology of industrial data?

Brainstorming (5 min)

List the industrial data and classify them into classes / types

towards a data typology

By nature of the industrial signal

1. Real-time data (streaming)
2. Event data
3. Transactional data (MES, ERP)
4. Contextual data / master data
5. Unstructured data (industrial)

By architecture layer (OT / IT)

1. OT level (Operational Technology)
2. MES level (industrial execution)
3. IT level (ERP / corporate)
4. Data platform level

By temporal granularity

- real time (milliseconds / seconds)
- near real time
- batch (hourly / daily)
- consolidated history

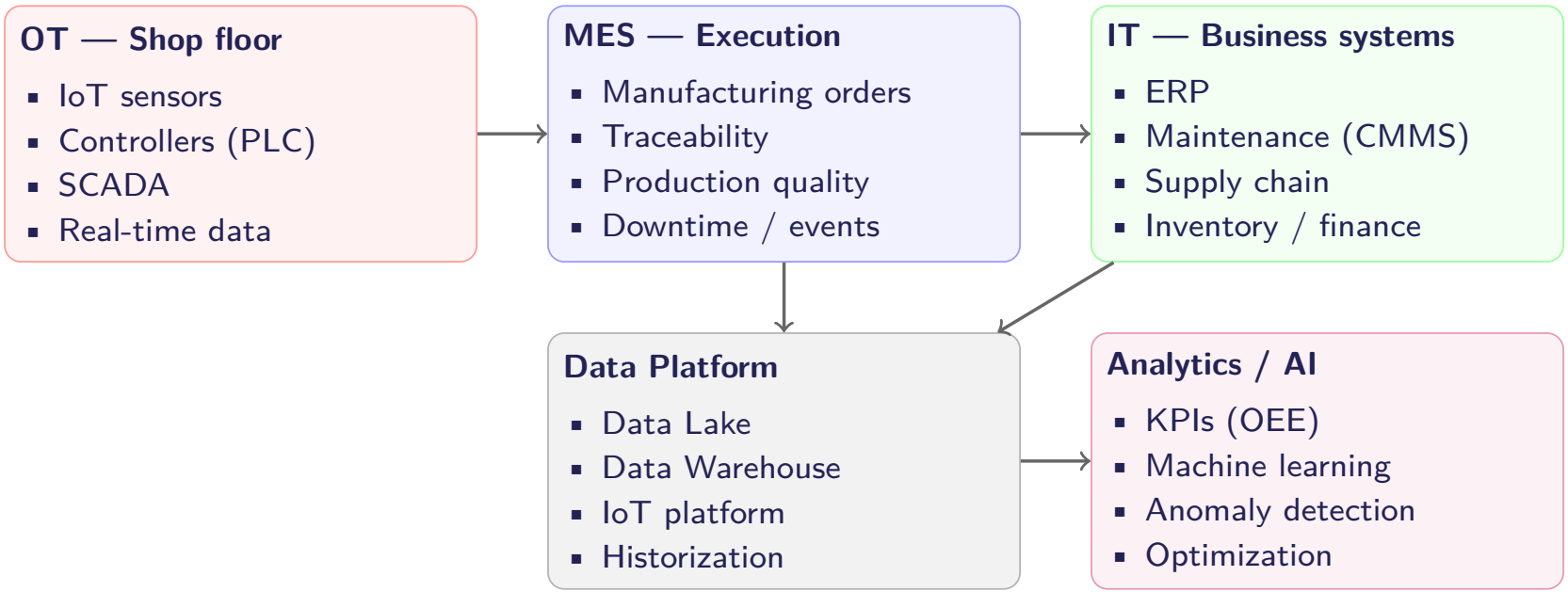
Analytical typology

- Continuous data
- Discrete data (events)
- Categorical data (codes, statuses)
- Derived data (KPIs)

The quality of industrial data depends strongly on its nature:

a sensor reading, an event and an ERP record have neither the same defects nor the same control mechanisms.

Industrial Data Landscape



Progressive transformation: *signal* $\rightarrow$ *information* $\rightarrow$ *decision*

The 6 pillars of data quality

1. Completeness

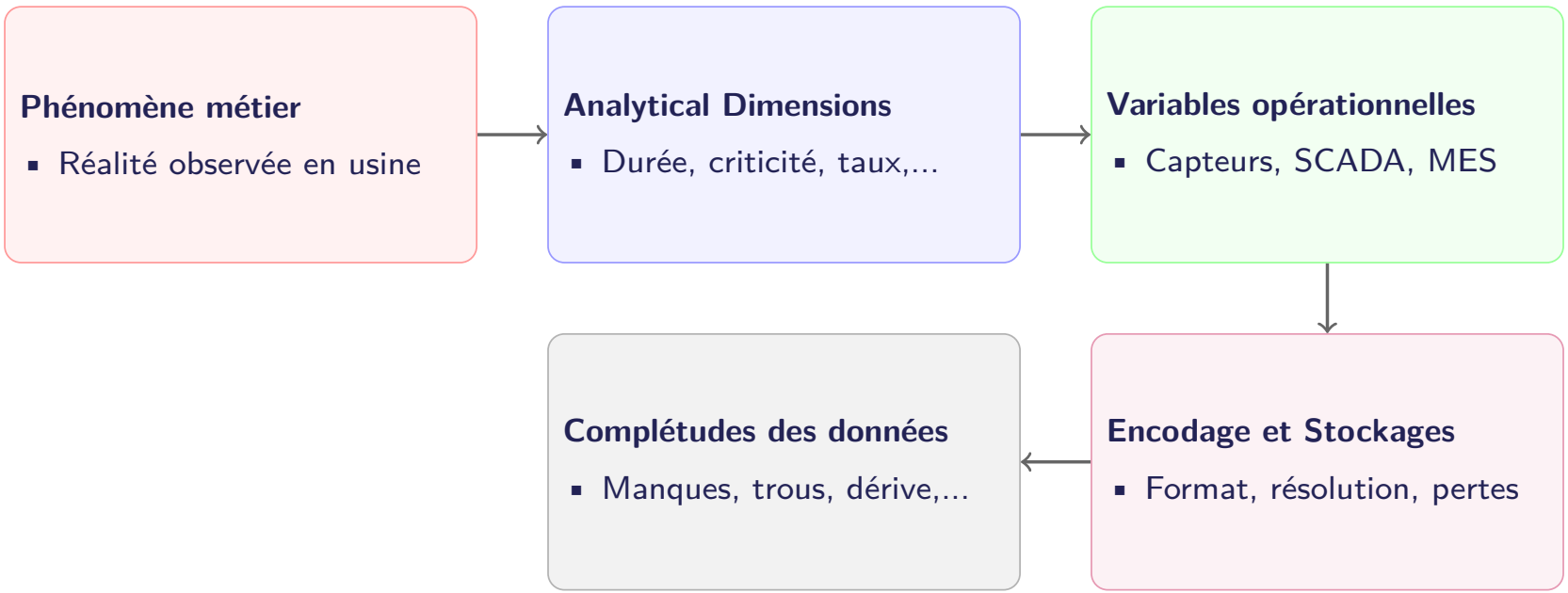
Is all the data present?

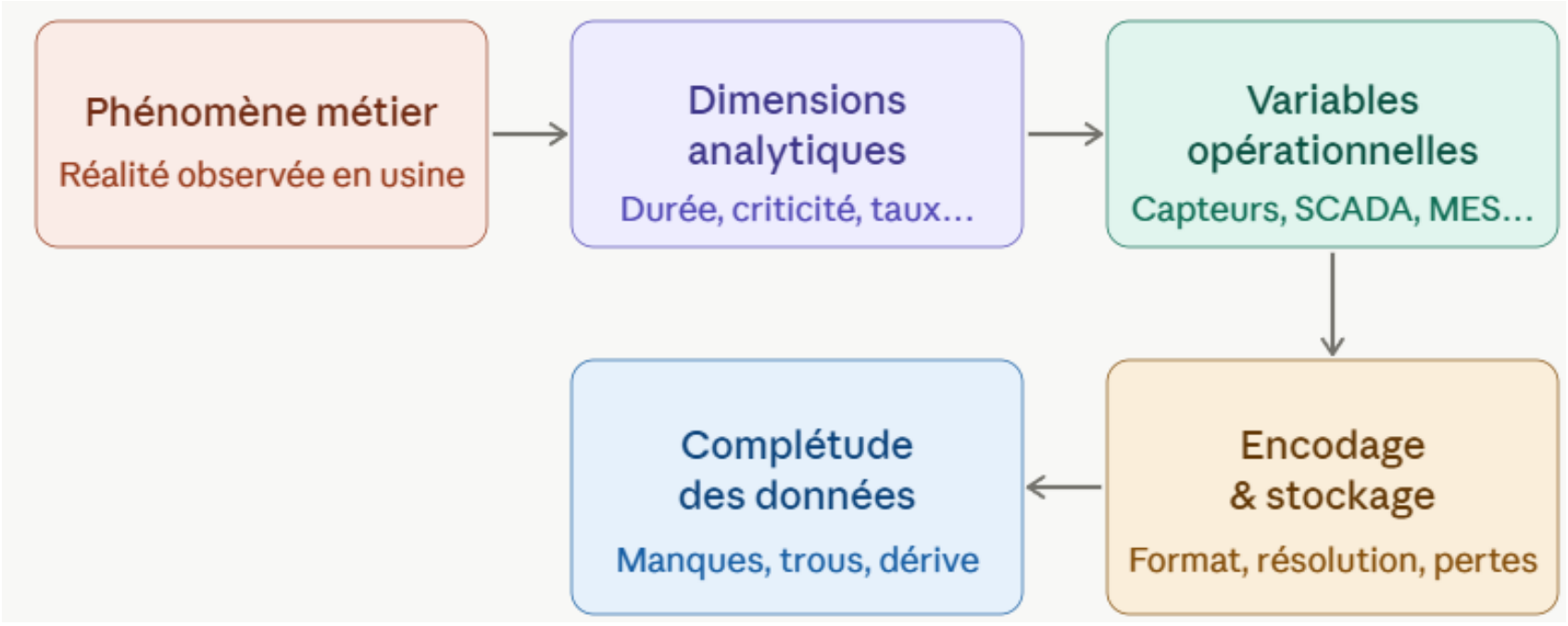
- Missing data (sensors, systems)
- Empty fields in industrial databases
- Collection interruptions

Indicators:

- rate of null values
- temporal coverage rate

1. Completeness




Phénomène métier
Réalité observée en usine
Dimensions analytiques
Durée, criticité, taux…
Variables opérationnelles
Capteurs, SCADA, MES…
Encodage & stockage
Format, résolution, pertes
Complétude des données
Manques, trous, dérive

1) Business phenomenon

→ what exists in the factory

2) Analytical dimensions

→ how we conceptualize the phenomenon

3) Operational variables

→ how it is actually measured in the systems

4) Computerized data

→ how it is stored and encoded

5) Completeness

→ do these measurements exist, correctly and without gaps?

6) Data preparation

→ Prepare the data for integration into a model

2. Accuracy

Does the data reflect reality?

- poorly calibrated sensors
- measurement drift
- human errors

Example: Measured temperature $\neq$ actual temperature

3. Consistency

Is the data compatible across sources?

- ERP vs MES inconsistencies
- different units
- logical contradictions

Example: Declared production > machine capacity

4. Timeliness

Is the data synchronized?

- IoT transmission latency
- incorrect timestamps
- misalignment between systems

Impact: distorted analysis of industrial performance

5. Uniqueness

Are there duplicates?

- sensor duplication
- ETL errors (extract/tranform/load)
- data resubmission

Effect: overestimation of production or incidents

6. Validity

Does the data comply with business rules?

- out-of-range values
- incorrect formats
- non-compliance with industrial rules

Example: negative pressure is impossible

Diagnostic methodology

Key steps

1. Map the data sources
2. Define the quality rules
3. Measure the indicators (KPIs)
4. Identify the root causes

Diagnostic tools

- Data profiling
- Data observability
- Real-time monitoring
- Quality dashboards

Conclusion

The 6 pillars of industrial data quality:

- Completeness
- Accuracy
- Consistency
- Timeliness
- Uniqueness
- Validity

The value of Industry 4.0 depends directly on trust in its data.

It is not the quantity of data that matters, but its reliability.

Data Vizu

Preliminary to data analysis and vizu

- Raw, unprocessed data is often not ready for visualisation.
- Techniques to turn data into information including:

Derivation and feature extraction

- Derivation and feature extraction: to add to existing data without requiring external datasets.
- additional columns *or features* based upon the existing data

Derived data element: using a mathematical, logical or other type of transformation, e.g. arithmetic formula, composition, aggregation.

Feature extraction: no math function, but a rule.

- a person in a family / a city from list of adresses

Combining datasets

- **Merging or Combining** two datasets with 10 columns each will be one dataset with 19 columns.
- Merging data can only be done with a shared value (jointure key)

Enriching geographic data

- Combine geographic data based upon location: spatial join.
- Essential stage in preparing geographic data for analysis is geocoding
- Geocoding is the process of taking any reference or description of a physical location and adding the actual physical location coordinates to the data.

Qualitative and quantitative data vizu

- qualitative research deals with open-ended, often text-based data
  - Barplot, surface plot, ...
- quantitative research tries to focus on objective, measurable data in the form of numbers or other structured data.
  - boxplot, histogram, scatter plot

The goals of visualising data

Communicate information clearly and efficiently to users through graphical means

- Aesthetic form and functionality to communicate its key-aspects of a dataset
- Communication is only one goal of a data visualisation
- Data visualisations as a support during the data analysis stage

Visualization and perception

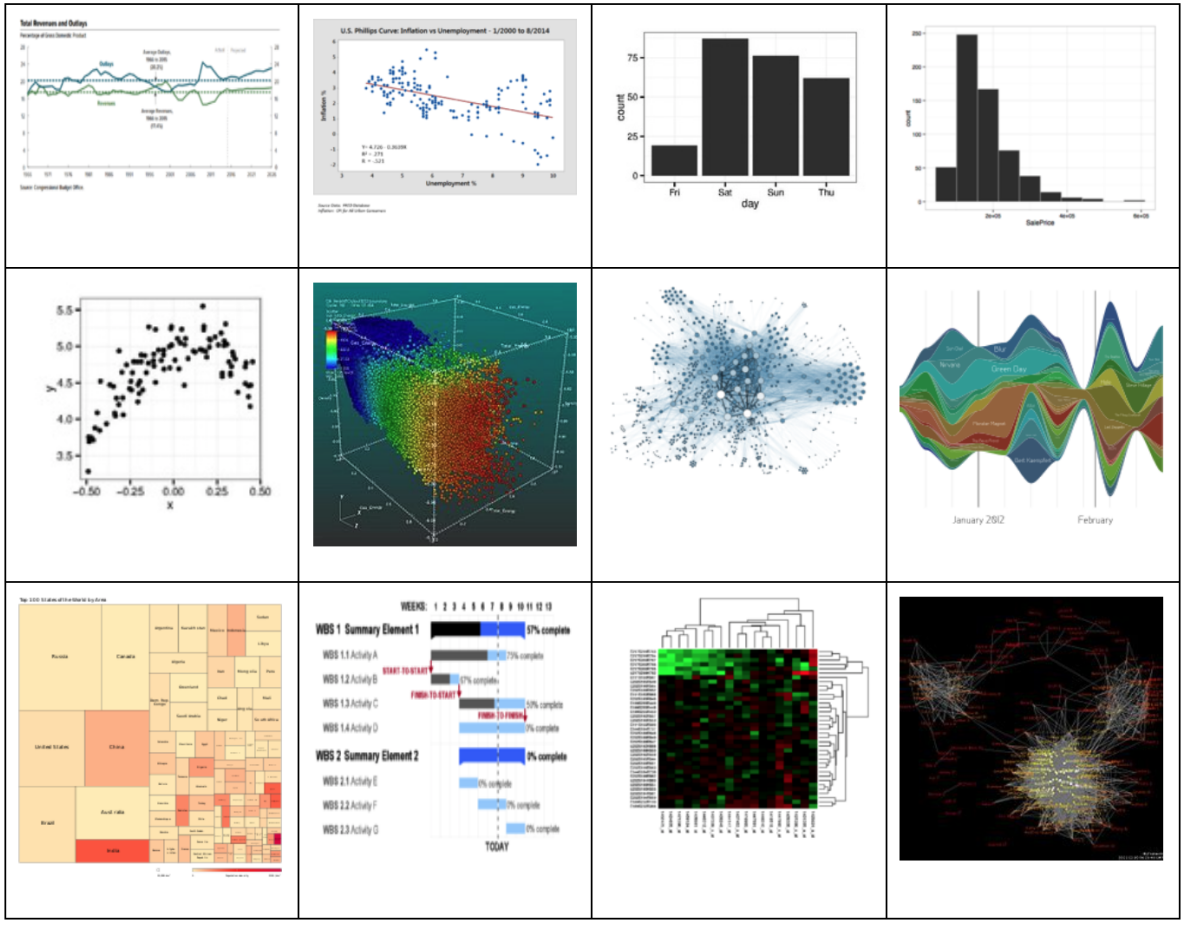


Figure 2: Examples of data visualisation *from Wikipedia*

Human eye is drawn towards

- brighter colours
- larger items
- things which stand out through difference.

The pop-out effect - Inside the visual cortex

- **dorsal** stream: *where/how*
  - information about our surroundings in real time
  - where things are and how they relate to other things
- **ventral** stream: *what*
  - 'what' the thing is
  - much slower process

Best data visualisations appeal to the dorsal stream and make information pop

Visualization and colours

Colour makes information ‘pop out’ also convey meaning

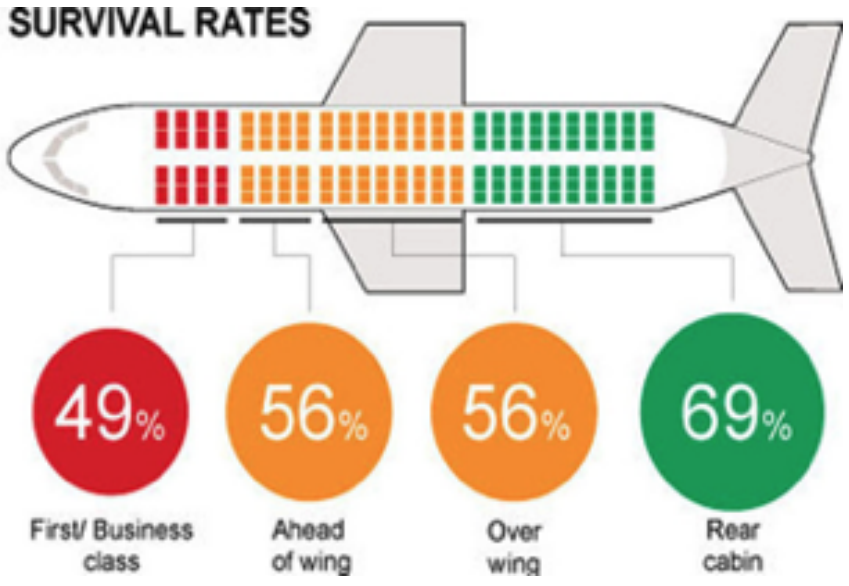


Figure 3: Traffic-light colours indicating safety

keep the number of colours used to a minimum

Choosing the correct visualisation for data

Choosing the correct visualisation for data depends on two key aspects:

- the type of data
- the message to be conveyed.

Data type - Description - Example visualisation

- Time series
  - Observations of the same objects over time
  - Line chart, motion chart, polar area diagram, Gantt chart, bar chart
- Population
  - Observation of different objects at a single point in time
  - Bar chart, map, treemap, pie chart
- Multivariate
  - Observations of different objects at different points in time
  - Multidimensional motion chart, bar chart, treemap |

Visualization

Visualization

> *Computer-based visualization systems provide visual representations of datasets designed to help people carry out tasks more effectively. (Munzner (2015))*

Visualization answers the need to augment human capabilities

- Exploration to rise questions
- Presentation of known results
- Represent understanding of requirements before developing models
- help developers of automatic solution refine/debug, determine parameters
- help end users of automatic solutions verify, build trust

Aid to analysis

Summaries lose information, details matter

- confirm expected and find unexpected patterns
- assess validity of statistical model

Framework

Munzner (2015) proposes a framework

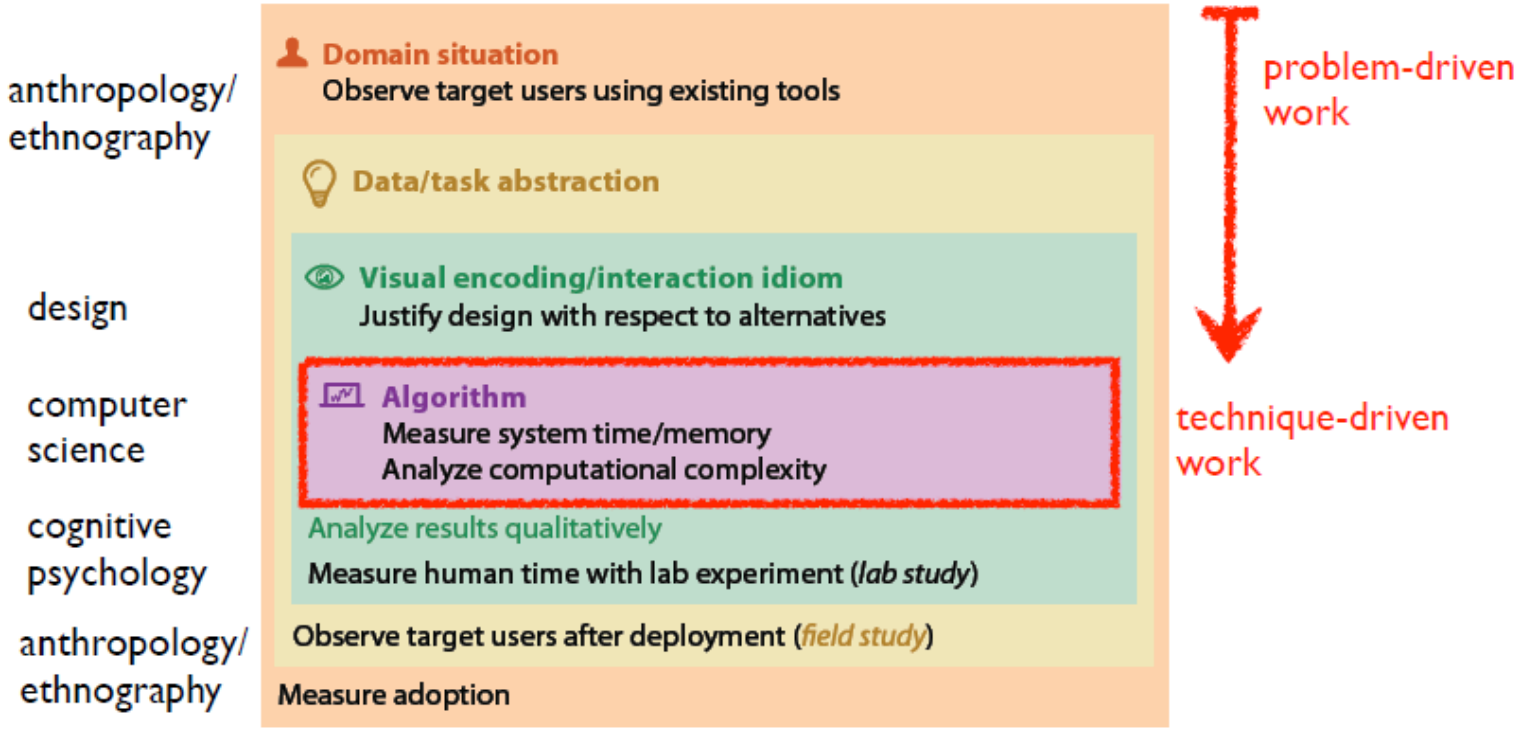


Figure 4: Munzner Framework

What? Why? How?

What can be visualized: data, datasets, and attributes

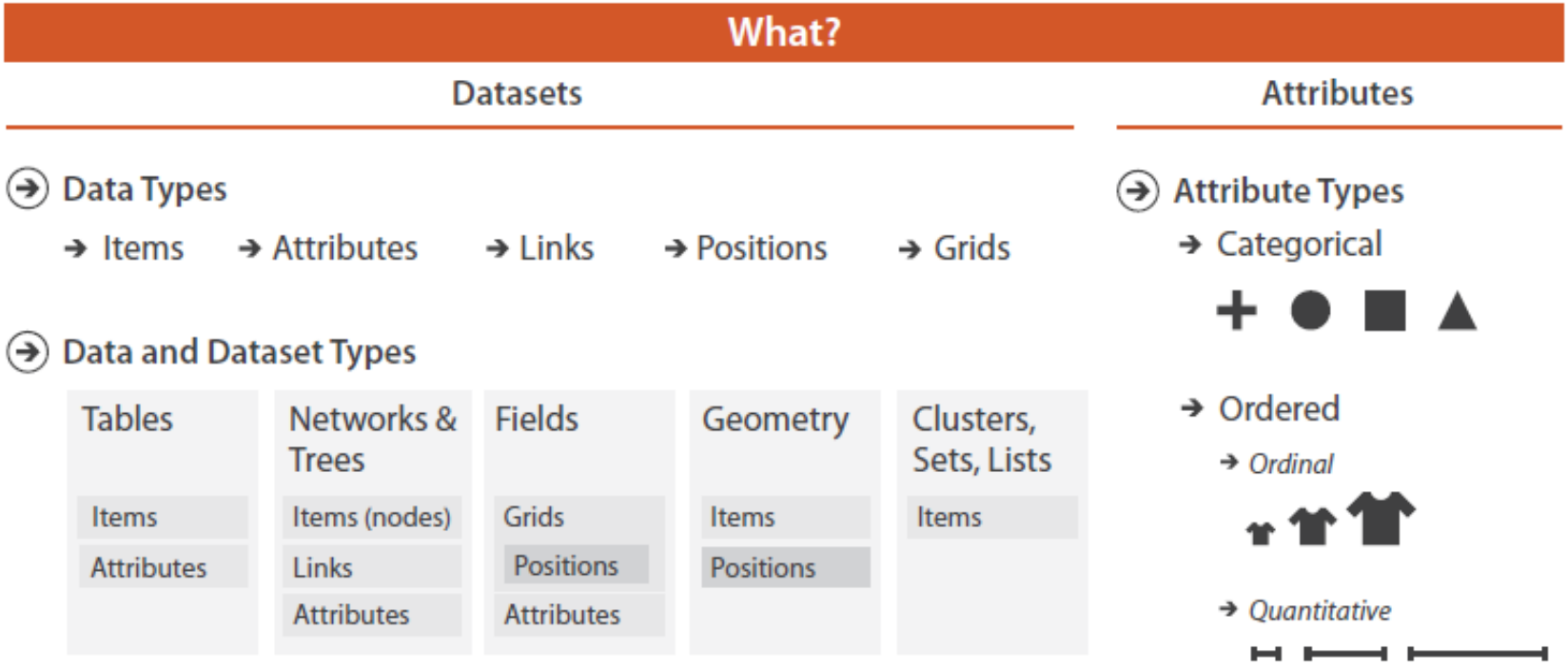
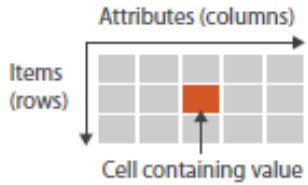


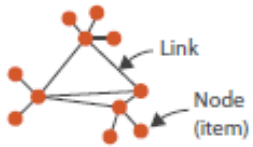
Figure 5: What - 1

Dataset Types

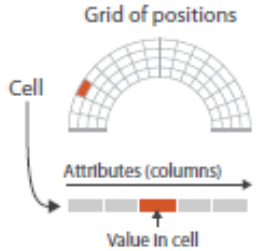
Tables



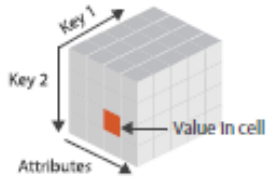
Networks



Fields (Continuous)



Multidimensional Table



Trees



Geometry (Spatial)



Ordering Direction

Sequential



Diverging



Cyclic



Dataset Availability

Static



Dynamic





Figure 6: What - 2

Why people are using vis in terms of actions and targets

{action, target} pairs:

- discover distribution
- compare trends
- locate outliers
- browse topology


Why?
Actions
Targets
Analyze
Consume
Discover
Present
Enjoy
Produce
Annotate
tag
Record
Derive
Search
Target known
Target unknown
Location known
Lookup
Browse
Location unknown
Locate
Explore
Query
Identify
Compare
Summarize
All Data
Trends
Outliers
Features
Attributes
One
Many
Distribution
Dependency
Correlation
Similarity
Extremes
Network Data
Topology
Paths
Spatial Data
Shape
What?
Why?
How?

Figure 7: Why

How to design vis idioms: encode, manipulate, facet, and reduce.

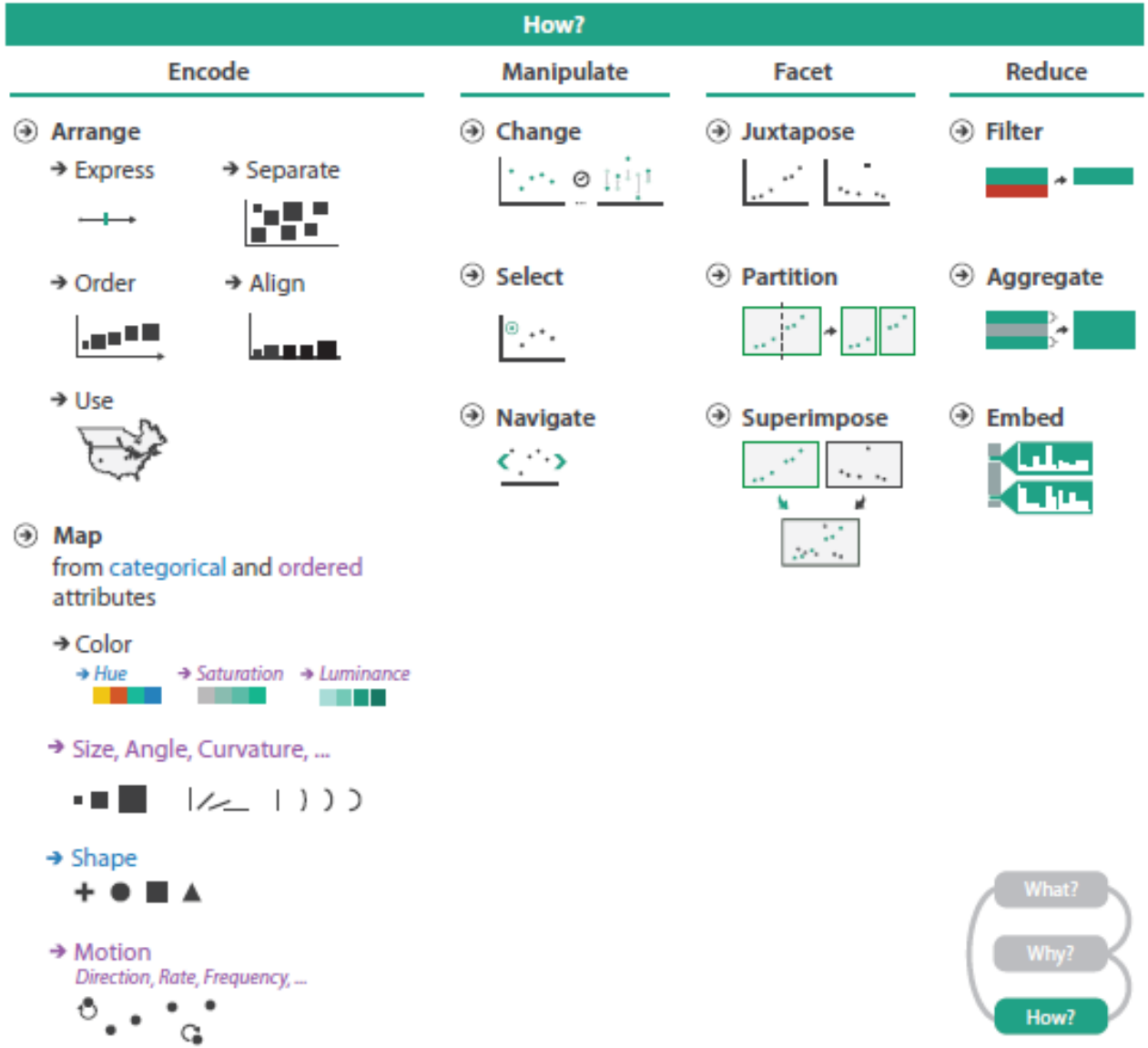


Figure 8: How

Eight rules of thumb.

- No Unjustified 3D
  - The Power of the Plane
  - The Disparity of Depth
  - Occlusion Hides Information
  - Perspective Distortion Dangers
  - Tilted Text Isn't Legible
- No Unjustified 2D
- Eyes Beat Memory
- Resolution over Immersion
- Overview First, Zoom and Filter, Detail on Demand
- Responsiveness Is Required
- Get It Right in Black and White
- Function First, Form Next

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References

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